

APCL

Ambiguity-Preserving Contact Localization

Preserve ambiguity. Move to resolve it.

The idea

A single arm configuration leaves contact location ambiguous along the force line of action. APCL preserves contact hypotheses along the unresolved direction and selects a new orientation to obtain complementary information. It uses joint states and commanded torques, without an object model or joint-torque, force/torque, or tactile measurements.

Available simulation evidence

A paired recovery study on a MuJoCo Franka FR3: 1,200 trials, adaptive stopping at 2–5 views. These figures use the full study before trust-gate filtering.

Method	Median	95th percentile	CBW	95% coverage
CPF / no recovery	9.2 mm	137.6 mm	17.67%	68.5%
APCL	6.6 mm	21.9 mm	0.83%	81.9%

CBW: reported 95% radius < 10 mm and true position error > 20 mm. Empirical coverage is below the nominal 95% level; uncertainty calibration remains incomplete.

What the film shows

Development seed 17, rerun for two views: 166.3 mm error without recovery and 3.2 mm with APCL. The film replays the recorded APCL joint trajectory and saved posterior snapshots. Each variant selects its own next orientation. The trajectory is slowed and includes holds; particle motion between saved snapshots is not synthesized.

Scope and limitations

The simulation uses a rigidly grasped payload, one object-fixed contact point, an approximately world-fixed applied force, quasi-static measurement windows, and regulated tool position. It does not model a physical pusher in the replay.

A batch re-estimate on the same measurement windows can match filter accuracy and yield better-calibrated uncertainty. APCL is not established as universally more accurate or faster. Hardware validation is pending. This note is not a submitted or accepted journal article.

Provenance

Source: `claude_try/results/main.jsonl` (1,200 trials).

SHA-256: `f5028492078b81dde049b0368eb7ecc0`

`2fbe2085f148dcda7c36c8bb13f4a45c`

The companion data archive contains per-trial CSV records, episode trajectories, weighted particle snapshots, metadata, and source hashes.